4 The partition function

Our last illustration of the techniques developed in this appendix is in their application to the partition function $p(n)$, which was discussed in Chapter 10. We derive for it the main term of the remarkable asymptotic formula of Hardy-Ramanujan.

Theorem 4.1 *If p denotes the partition function, then*

(i) $p(n) \sim \dfrac{1}{4\sqrt{3}n}\, e^{Kn^{1/2}}$ *as $n \to \infty$, where $K = \pi\sqrt{\frac{2}{3}}$.*

(ii) *A much more precise assertion is that*

$$p(n) = \frac{1}{2\pi\sqrt{2}}\frac{d}{dn}\left(\frac{e^{K(n-\frac{1}{24})^{1/2}}}{(n-\frac{1}{24})^{1/2}}\right) + O(e^{\frac{K}{2}n^{1/2}}).$$

Note. Observe that $(n-\frac{1}{24})^{1/2} - n^{1/2} = O(n^{-1/2})$, by the mean-value theorem; hence $e^{K(n-\frac{1}{24})^{1/2}} = e^{Kn^{1/2}}(1+O(n^{-1/2}))$, thus $e^{K(n-\frac{1}{24})^{1/2}} \sim e^{Kn^{1/2}}$, as $n \to \infty$. Of course, clearly $(n-\frac{1}{24})^{1/2} \sim n^{1/2}$, and in particular (ii) implies (i).

We shall discuss first, in a more general setting, how we might derive the asymptotic behavior of a sequence $\{F_n\}$ from the analytic properties of its generating function $F(w) = \sum_{n=0}^{\infty} F_n w^n$. Assuming for the sake of simplicity that $\sum F_n w^n$ has the unit disc as its disc of convergence, we can set forth the following heuristic principle: the asymptotic behavior of F_n is determined by the location and nature of the "singularities" of F on the unit circle, and the contribution to the asymptotic formula due to each singularity corresponds in magnitude to the "order" of that singularity.

A very simple example in which this principle is unambiguous and can be verified occurs when F is meromorphic in a larger disc, but has only one singularity on the circle, a pole of order r at the point $w = 1$. Then there is a polynomial P of degree $r-1$ so that $F_n = P(n) + O(e^{-\epsilon n})$ as $n \to \infty$, for some $\epsilon > 0$. In fact, $\sum_{n=0}^{\infty} P(n)w^n$ is a good approximation to $F(w)$ near $w = 1$; it is the principal part of the pole of F. (See also Problem 4.)

For the partition function the analysis is not as simple as this example, but the principle stated above is still applicable when properly interpreted. To this task we now turn.

We recall the formula

$$\sum_{n=0}^{\infty} p(n)\, w^n = \prod_{n=1}^{\infty} \frac{1}{1-w^n},$$

established in Theorem 2.1, Chapter 10. This identity implies that the generating function is holomorphic in the unit disc. In what follows, it will be convenient to pass from the unit disc to the upper half-plane by writing $w = e^{2\pi iz}$, $z = x + iy$, and taking $y > 0$. We therefore have

$$\sum_{n=0}^{\infty} p(n)e^{2\pi inz} = f(z),$$

with

$$f(z) = \prod_{n=1}^{\infty} \frac{1}{1 - e^{2\pi inz}},$$

and

$$p(n) = \int_\gamma f(z)e^{-2\pi inz}\, dz. \tag{23}$$

Here γ is the segment in the upper half-plane joining $-1/2 + i\delta$ to $1/2 + i\delta$, with $\delta > 0$; the height δ will be fixed later in terms of n.

To proceed further, we look first at where the main contribution to the integral (23) might be, in terms of the relative size of $f(x+iy)$, as $y \to 0$. Notice that f is largest near $z = 0$. This is because $|f(x+iy)| \leq f(iy)$, and moreover $f(iy)$ increases as y decreases, in view of the fact that the coefficients $p(n)$ are positive. Alternatively, we observe that each factor $1 - e^{2\pi inz}$, appearing in the product for f, vanishes as $z \to 0$, but the same is true for any other point (mod 1) on the real axis. Thus in analogy with the simple example considered above, we seek an elementary function f_1, which has much the same behavior as f at $z = 0$, and try to replace f by f_1 in (23).

It is here that we are very fortunate, because the generating function is just a variant of the Dedekind eta function,

$$\eta(z) = e^{i\pi z/12} \prod_{n=1}^{\infty} (1 - e^{2\pi inz}).$$

From this, it is obvious that

$$f(z) = e^{\frac{i\pi z}{12}} (\eta(z))^{-1}.$$

(Incidentally, the fraction $1/12$ arising above will explain the occurrence of the fraction $1/24$ in the asymptotic formula for $p(n)$.)

Since η satisfies the functional equation $\eta(-1/z) = \sqrt{z/i}\ \eta(z)$ (see Proposition 1.9 in Chapter 10), it follows that

$$f(z) = \sqrt{z/i}\ e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}} f(-1/z). \tag{24}$$

Notice also that if z is appropriately restricted and $z \to 0$, then $\mathrm{Im}(-1/z) \to \infty$, from which it follows that $f(-1/z) \to 1$ rapidly, because

$$f(z) = 1 + O(e^{-2\pi y}), \qquad z = x + iy,\ y \geq 1. \tag{25}$$

Thus it is natural to choose $f_1(z) = \sqrt{z/i}\ e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}}$ as the function which approximates well the generating function $f(z)$ (at $z = 0$), and write (because of (24))

$$p(n) = p_1(n) + E(n),$$

with

$$\begin{cases} p_1(n) &= \displaystyle\int_\gamma \sqrt{z/i}\ e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}} e^{-2\pi i n z}\, dz, \\ \\ E(n) &= \displaystyle\int_\gamma \sqrt{z/i}\ e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}} e^{-2\pi i n z} (f(-1/z) - 1)\, dz. \end{cases}$$

We first take care of the error term $E(n)$, and in doing so we specify γ by choosing its height in terms of n. In estimating $E(n)$ we replace its integrand by its absolute value and note that if $z \in \gamma$, then

$$\left| \sqrt{z/i}\ e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}} e^{-2\pi i n z} \right| \leq c e^{2\pi n \delta} e^{\frac{\pi}{12}\frac{\delta}{\delta^2 + x^2}}, \tag{26}$$

since $z = x + iy$, and $\mathrm{Re}(i/z) = \delta/(\delta^2 + x^2)$.

On the other hand, we can make two estimates for $f(-1/z) - 1$. The first arises from (25) by replacing z by $-1/z$, and gives

$$|f(-1/z) - 1| \leq c e^{-2\pi \frac{\delta}{\delta^2 + x^2}} \qquad \text{if } \tfrac{\delta}{\delta^2 + x^2} \geq 1. \tag{27}$$

For the second, we observe that $|f(z)| \leq f(iy) \leq C e^{\frac{\pi}{12y}}$, when $y \leq 1$, because of the functional equation (24), and hence

$$|f(-1/z) - 1| \leq O\left(e^{\frac{\pi}{12}\frac{\delta^2 + x^2}{\delta}}\right) = O\left(e^{\frac{\pi}{48\delta}}\right) \tag{28}$$

if $\frac{\delta}{\delta^2 + x^2} \leq 1$, since $|x| \leq 1/2$.

Therefore in the integral defining $E(n)$ we use (26) and (27) when $\frac{\delta}{\delta^2+x^2} \geq 1$, and (26) and (28) when $\frac{\delta}{\delta^2+x^2} \leq 1$. The first leads to a contribution of $O(e^{2\pi n\delta})$, since $2\pi > \pi/12$. The second gives a contribution of $O(e^{2\pi n\delta} e^{\frac{\pi}{48\delta}})$. Hence $E(n) = O(e^{2\pi n\delta} e^{\frac{\pi}{48\delta}})$, and we choose δ so as to minimize the right-hand side, that is, $2\pi n\delta = \frac{\pi}{48\delta}$; this means we take $\delta = \frac{1}{4\sqrt{6}\, n^{1/2}}$, and we get

$$E(n) = O\left(e^{\frac{4\pi}{4\sqrt{6}}\, n^{1/2}}\right) = O\left(e^{\frac{K}{2}\, n^{1/2}}\right),$$

which is the desired size of the error term.

We turn to the main term $p_1(n)$. To simplify later calculations we "improve" the contour γ by adding to it two small end-segments; these are the segment joining $-1/2$ to $-1/2 + i\delta$ and that joining $1/2 + i\delta$ to $1/2$. We call this new contour γ' (see Figure 5).

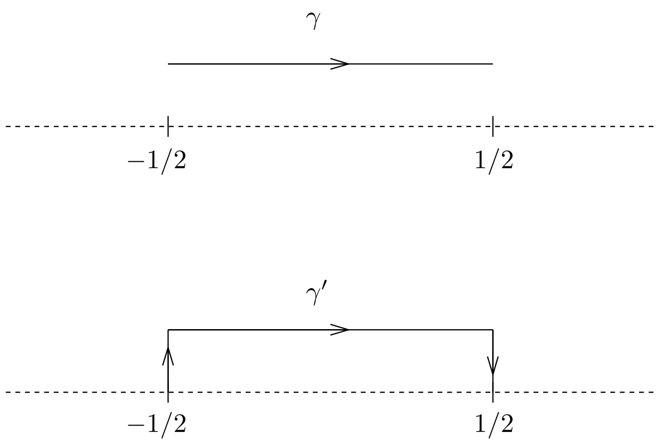

Figure 5. γ and the improved contour γ'

Notice that since $\sqrt{z/i}\; e^{\frac{i\pi}{12z}}$ is $O(1)$ on the two added segments (for the integral defining $p_1(n)$), the modification contributes $O(e^{2\pi n\delta}) = O(e^{\frac{2\pi}{4\sqrt{6}} n^{1/2}}) = O(e^{\frac{K}{4} n^{1/2}})$, which is even smaller than the allowed error, and therefore can be incorporated in $E(n)$. So without introducing further notation we will rewrite $p_1(n)$ replacing the contour γ by γ' in the integration defining p_1, namely

$$p_1(n) = \int_{\gamma'} \sqrt{z/i}\; e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}} e^{-2\pi i n z}\, dz. \tag{29}$$

Next we simplify the triad of exponentials appearing in (29) by making a change of variables $z \mapsto \mu z$ so that their combination takes the form

$$e^{Ai\left(\frac{1}{z}-z\right)}.$$

This can be achieved under the two conditions $A = 2\pi\mu(n - \frac{1}{24})$ and $A = \frac{\pi}{12\mu}$, which means that

$$A = \frac{\pi}{\sqrt{6}}\,(n - \frac{1}{24})^{1/2} \quad \text{and} \quad \mu = \frac{1}{2\sqrt{6}}\,(n - \frac{1}{24})^{-1/2}.$$

Making the indicated change of variables we now have

$$p_1(n) = \mu^{3/2} \int_\Gamma e^{-sF(z)}\ \sqrt{z/i}\ dz, \tag{30}$$

with $F(z) = i(z - 1/z)$, $s = \frac{\pi}{\sqrt{6}}\,(n - \frac{1}{24})^{1/2}$. The curve Γ (see Figure 6). is now the union of three segments $[-a_n, -a_n + i\delta']$, $[-a_n + i\delta', a_n + i\delta']$, and $[a_n + i\delta', a_n]$; we can write $\Gamma = \mu^{-1}\gamma'$.

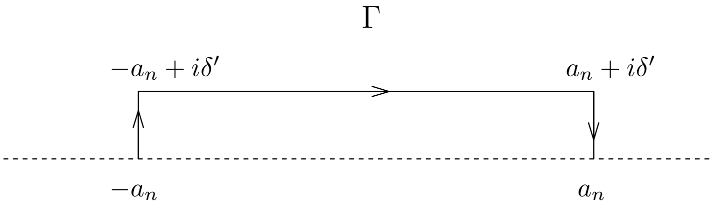

Figure 6. The curve Γ

Here $a_n = \frac{1}{2}\mu^{-1} = \sqrt{6}\,(n - \frac{1}{24})^{1/2} \approx n^{1/2}$, while $\delta' = \delta\mu^{-1} = \frac{2\sqrt{6}}{4\sqrt{6}n^{1/2}}\,(n - \frac{1}{24})^{1/2} \sim 1/2$, as $n \to \infty$.

We apply the method of steepest descent to the integral (30). In doing this, we note that $F(z) = i(z - 1/z)$ has one (complex) critical point $z = i$, in the upper half-plane. Moreover, the two curves passing through i on which F is real are: the imaginary axis, on which F has a maximum at $z = i$, which we reject, and the unit circle, on which F has a minimum at $z = i$. Thus using Cauchy's theorem we replace the integration on Γ by the integration over our final curve Γ^*, which consists of the segment $[-a_n, -1]$, $[1, a_n]$, together with the upper semicircle joining -1 to 1.

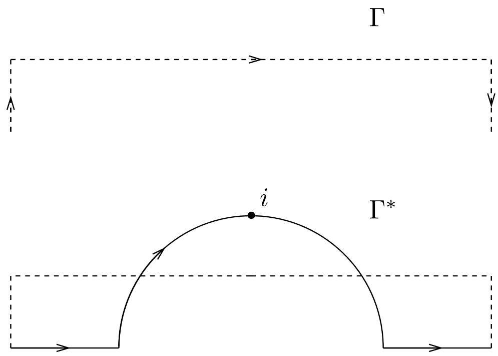

Figure 7. The final curve Γ^*

We therefore have

$$p_1(n) = \mu^{3/2} \int_{\Gamma^*} e^{-sF(z)} \sqrt{z/i}\ dz.$$

The contributions on the segments $[-a_n, -1]$ and $[1, a_n]$ are relatively very small, because on the real axis the exponential has absolute value 1, and hence the integrand is bounded by $\sup_{|z|\le a_n} |z|^{1/2}$, and this leads to two terms which are $O(a_n^{3/2}\mu^{3/2}) = O(1)$.

Finally, we come to the principal part, which is the integration over the semicircle, taken with the orientation on the figure. Here we write $z = e^{i\theta}$, $dz = ie^{i\theta}\, d\theta$. Since $i(z - 1/z) = -2\sin\theta$, this gives a contribution

$$-\mu^{3/2}\int_0^{\pi} e^{2s\sin\theta} e^{i3\theta/2}\, \sqrt{i}\, d\theta = \mu^{3/2}\int_{-\pi/2}^{\pi/2} e^{2s\cos\theta}(\cos(3\theta/2) + i\sin(3\theta/2))d\theta.$$

In applying Proposition 2.1, Laplace's method, we take $\Phi(\theta) = -\cos\theta$, $\theta_0 = 0$, so $\Phi(\theta_0) = -1$, $\Phi''(\theta_0) = 1$ and we choose $\psi(\theta) = \cos(3\theta/2) + i\sin(3\theta/2)$, so that $\psi(\theta_0) = 1$. Therefore, the above contributes

$$\mu^{3/2} e^{2s} \frac{\sqrt{2\pi}}{(2s)^{1/2}} \left(1 + O(s^{-1/2})\right).$$

Now since $s = \frac{\pi}{\sqrt{6}}\,(n - \frac{1}{24})^{1/2}$, $\frac{2\pi}{\sqrt{6}} = \pi\sqrt{\frac{2}{3}} = K$, and $\mu = \frac{\sqrt{6}}{12}\,(n - \frac{1}{24})^{-1/2}$, we obtain

$$p(n) = \frac{1}{4n\sqrt{3}}\, e^{K\, n^{1/2}} \left(1 + O\left(n^{-1/4}\right)\right),$$

and the first conclusion of the theorem is established.

To obtain the more exact conclusion (ii), we retrace our steps and use an additional device, which allows us to evaluate rather precisely the key integral. With $p_1(n)$ defined by (29), which is an integral taken over $\gamma' = \gamma'_n$, we write

$$p_1(n) = \frac{d}{dn}q(n) + e(n),$$

where

$$q(n) = \frac{1}{2\pi}\int_{\gamma'} (z/i)^{-1/2} e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}} e^{-2\pi i n z}\, dz,$$

and $e(n)$ is the term due to the variation of the contour $\gamma' = \gamma'_n$, when forming the derivative in n. By Cauchy's theorem this is easily seen to be dominated by $O(e^{2\pi n\delta})$, which we have seen is $O(e^{\frac{K}{4}n^{1/2}})$, and can be subsumed in the error term. To analyze $q(n)$, we proceed as before, first making the change of variables $z \mapsto \mu z$, and then replacing the resulting contour Γ by Γ^*. As a consequence, we have

$$q(n) = \frac{\mu^{1/2}}{2\pi}\int_{\Gamma^*} e^{-sF(z)}(z/i)^{-1/2}\, dz, \tag{31}$$

with $F(z) = i(z - 1/z)$, $s = \frac{\pi}{\sqrt{6}}(n - \frac{1}{24})^{1/2}$, and $\mu = \frac{1}{2\sqrt{6}}(n - \frac{1}{24})^{-1/2}$.

Now the two segments $[-a_n, -1]$ and $[1, a_n]$ of the contour Γ^* make harmless contributions to $\frac{d}{dn}q(n)$, since F is purely imaginary on the real axis. Indeed, they yield terms which are $O(a_n^{1/2}\mu^{1/2}) = O(1)$.

The main part of (31) is the term arising from the integration on the semicircle. Thus setting $z = e^{i\theta}$, $dz = ie^{i\theta}\, d\theta$, and $i(z - 1/z) = -2\sin\theta$, it equals

$$\begin{aligned}
-\frac{\mu^{1/2}}{2\pi}\int_0^{\pi} e^{2s\sin\theta} e^{i\theta/2} i^{3/2}\, d\theta &= \frac{\mu^{1/2}}{2\pi}\int_{-\pi/2}^{\pi/2} e^{2s\cos\theta}(\cos(\theta/2) + i\sin(\theta/2))d\theta \\
&= \frac{\mu^{1/2}}{2\pi}\int_{-\pi/2}^{\pi/2} e^{2s\cos\theta}\cos(\theta/2)d\theta,
\end{aligned}$$

where we have used the fact that the integral $\int_{-\pi/2}^{\pi/2} e^{2s\cos\theta}\sin(\theta/2)\, d\theta$ vanishes since the integrand is odd.

Now $\cos\theta = 1 - 2(\sin\theta/2)^2$, so setting $x = \sin(\theta/2)$ we see that the above integral becomes

$$\frac{\mu^{1/2}e^{2s}}{\pi}\int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} e^{-4sx^2}\,dx.$$

However

$$\begin{aligned}\int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} e^{-4sx^2}\,dx &= \int_{-\infty}^{\infty} e^{-4sx^2}\,dx + O\left(\int_{\frac{\sqrt{2}}{2}}^{\infty} e^{-4sx^2}\,dx\right)\\ &= \frac{\sqrt{\pi}}{2s^{1/2}} + O(e^{-2s}),\end{aligned}$$

and also

$$\frac{d}{ds}\left(\int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} e^{-4sx^2}\,dx\right) = \frac{d}{ds}\left(\frac{\sqrt{\pi}}{2s^{1/2}}\right) + O(e^{-2s}).$$

Gathering all the error terms together, we find

$$p(n) = \frac{d}{dn}\left(\mu^{1/2}\frac{e^{2s}}{\pi}\frac{\sqrt{\pi}}{2s^{1/2}}\right) + O(e^{\frac{K}{2}n^{1/2}}).$$

Since $s = \frac{\pi}{\sqrt{6}}(n - \frac{1}{24})^{1/2}$, $\mu = \frac{\sqrt{6}}{12}(n - \frac{1}{24})^{-1/2}$, and $K = \pi\sqrt{\frac{2}{3}}$, this is

$$p(n) = \frac{1}{2\pi\sqrt{2}}\frac{d}{dn}\left(\frac{e^{K(n-\frac{1}{24})^{1/2}}}{(n-\frac{1}{24})^{1/2}}\right) + O(e^{\frac{K}{2}n^{1/2}}),$$

and the theorem is proved.

5 Problems

1. Let η be an indefinitely differentiable function supported in a finite interval, so that $\eta(x) = 1$ for x near 0. Then, if $m > -1$ and $N > 0$,

$$\int_0^{\infty} e^{-sx}x^m\eta(x)\,dx = s^{-m-1}\Gamma(m+1) + O(s^{-N})$$

for $\mathrm{Re}(s) \geq 0$, $|s| \to \infty$.

(a) Consider first the case $-1 < m \leq 0$. It suffices to see that

$$\int_0^{\infty} e^{-sx}x^m(1-\eta(x))\,dx = O(s^{-N}),$$

and this can be done by repeated integration by parts since $e^{-sx} = (-1)^N s^{-N}\left(\frac{d}{dx}\right)^N(e^{-sx})$.